Preliminary

MEC165xB

Keyboard and Embedded Controller for Notebook PC

Operating Conditions

- Operating Voltages: 3.3 V and 1.8 V
- Operating Temperature Range: -40 ^{o}C to 85 ^{o}C

Low Power Modes

- Chip is designed to always operate in Lowest Power state during Normal Operation
- Supports all 5 ACPI Power States for PC platforms
- Supports 2 Chip-level Sleep Modes: Light Sleep and Heavy Sleep
  - Low Standby Current in Sleep Modes

ARM® Cortex-M4 Embedded Processor

- Programmable clock frequency up to 48 MHz
- Fixed point processor
- Single 4GByte Addressing Space
- Nested Vectored Interrupt Controller (NVIC)
  - Maskable Interrupt Controller
  - Maskable hardware wake up events
  - 8 Levels of priority, individually assignable by vector
- EC Interrupt Aggregator expands number of Interrupt sources supported or reduces number of vectors needed
- Complete ARM® Standard debug support
  - JTAG-Based DAP port, comprised of SWJ-DP and AHB-AP debugger access functions

Memory Components

- 416KB Code/Data SRAM
  - 352KB optimized for code performance
  - 64KB optimized for data performance
- 128 Bytes Battery Powered Storage SRAM
- 4K bits OTP
  - In circuit programmable
- ROM
- Contains Boot ROM
  - Contains Runtime APIs for built-in functions
  - 224KB of ROM space
- 8KB Internal EEPROM
- 512K Byte Internal SPI

Clocks

- 96 MHz Internal PLL
- 32 kHz Clock Sources
  - Internal 32 kHz silicon oscillator
  - External 32 kHz crystal (XTAL) source
  - External single-ended 32 kHz clock source

Package Options

- 128 pin WFBGA
- 144 pin WFBGA

Security Features

- Boot ROM Secure Boot Loader
  - Hardware Root of Trust (RoT) using Secure Boot and Immutable code
  - Supports 2 Code Images in external SPI Flash (Primary and Fall back image)
  - Authenticates SPI Flash image before loading
  - Support AES-256 Encrypted SPI Flash images
  - SHA384 Integrity Check
  - Authentication with ECC and PQC Support
  - ECDSA-384 and/or PQC (LMS or ML-DSA)
  - LMS Secure Boot Parameter Set:
    - LMOTS_SHA256_N24_W4
    - LMS_SHA256_M24_H20
  - ML-DSA-87 (Crystals Dilithium-5)
  - Supports Transfer of Ownership
- Hardware Accelerators:
  - Multi purpose AES Crypto Engine:
    - Support for 128-bit - 256-bit key length
    - Supports Battery Authentication applications
  - Digital Signature Algorithm Support
    - Support for ECDSA and EC_KCDSA
  - Cryptographic Hash Engine
    - Support for SHA-1, SHA-256 to SHA-512
  - Public Key Crypto Engine
    - Hardware support for RSA and Elliptic Curve asymmetric public key algorithms
    - RSA keys length of 1024 to 4096 bits
    - ECC Prime Field keys up to 571 bits
    - ECC Binary Field keys up to 571 bits
    - Microcoded support for standard public key algorithms
  - OTP for storing Keys and IDs
    - Lockable on 32 B boundaries to prevent

read access or write access
- True Random Number Generator
- 1 Kbit FIFO
- JTAG Disabled by default

System Host interface

- Enhanced Serial Peripheral Interface (eSPI)
  - Intel eSPI Specification compliant
    - eSPI Interface Base Spec, Intel Doc. #327432-004, Rev. 1.0.
    - eSPI Compatibility Spec, Intel Doc. #562633, Rev. 0.6
  - ESPI Controller Attached Flash Sharing (CAFS)
  - ESPI Target Attached Flash Sharing (TAFS)
  - Supports all four channels:
    - Peripheral Channel
    - Virtual Wires Channel
    - Out-of-Band (OOB) Tunneled Message Channel
    - Run-time Flash Access Channel
  - Supports EC Bus Master to Host Memory
  - Supports up to 66 MHz maximum operating frequency
- RPMC Support
  - Support Two RPMC flashes
- System to EC Message Interface
  - Three Embedded Memory Interfaces
    - Provides Two Windows to On-Chip SRAM for Host Access
    - Two Register Mailbox Command Interface
  - Mailbox Registers Interface
    - Thirty-two 8-bit registers
    - Two Register Mailbox Command Interfaces
    - Two Register SMI Source Interfaces
  - Six ACPI Embedded Controller Interfaces
    - Five EC Interfaces
    - One Power Management Interface
- One Serial Peripheral Interface (SPI) Master Controller
  - Dual and Quad I/O Support
  - Flexible Clock Rates
  - Support for 1.8V and 3.3V slave devices
  - SPI Burst Capable
  - SPI Controller Operates with Internal DMA Controller with CRC Generation
  - Mappable to the following ports (only 1 port active at a time)
    - 1 shared SPI Interface2 General purpose SPI
    - 1 Private SPI Interface
    - 1 In-Chip SPI
- Two General purpose Serial Peripheral Interface (SPI) Controllers
  - One EC driven Full Duplex Serial Communication Interface
  - Flexible Clock Rates
- SPI burst capable8042 Emulated Keyboard Controller
  - 8042 Style Host Interface
  - Port 92 Legacy A20M Support
- Fast GATEA20 & Fast CPU_RESET18 x 8 Interrupt Capable Multiplexed Keyboard Scan Matrix
  - Optional Push-Pull Drive for Fast Signal Switching
- PECI Interface 3.1
  - Support Intel's low voltage PECI
- Port 80 BIOS Debug Port
  - Two Ports, Assignable to Any eSPI IO Address
  - 24-bit Timestamp with Adjustable Timebase
  - 16-Entry FIFO

Peripheral Features

- Internal DMA Controller
  - Hardware or Firmware Flow Control
  - Firmware Initiated Memory-to-Memory transfers
  - Hardware CRC-32 Generator on Channel 0
- 16-Hardware DMA Channels support five SMBus Master/Slave Controllers, One Quad SPI Controller and Two General purpose SPI Controllers I2C/SMBus Controllers
  - 5 I2C/SMBus controllers
  - 15 Configurable I2C ports
    - Full Crossbar switch allows any port to be connected to any controller
  - Supports Promiscuous mode of operation
  - Fully Operational on Standby Power
  - Multi-Master Capable
  - Supports Clock Stretching
  - Programmable Bus Speeds
  - 1 MHz Capable
  - Supports DMA Network Layer
- General Purpose I/O Pins
  - Inputs:
    - Asynchronous rising and falling edge wakeup detection Interrupt High or Low Level
  - Outputs:
    - Push Pull or Open Drain output
    - Programmable power well emulation
  - Pull up or pull down resistor control
    - Automatically disabling pull-up resistors when output driven low

  - Automatically disabling pull-down resistors when output driven high
- Programmable drive strength
- Two separate1.8V/3.3V configurable IO regions
- Group or individual control of GPIO data
- 13- Over voltage tolerant GPIO pins
- Glitch protection and Under-Voltage Protection on all GPIO pins
- 8 GPIO Pass through ports

• Input Capture and Compare timer
- Six 32-bit Capture Registers
- 16 Input Pins (ICTx)
  - Full Crossbar switch allows any port to be connected to any controller
- 32-bit Free-running timer
- Two 32-bit Compare Registers
- Capture, Compare and Overflow Interrupts

• Universal Asynchronous Receiver Transmitter (UART)
- Three High Speed NS16C550A Compatible UARTs with Send/Receive 16-Byte FIFOs
  -UART0 - Configurable 2-pin/4-pin/8-pin
  -UART1 - Configurable 2-pin/4-pin/8-pin
  -UART2 - Configurable 2-pin/4-pin/8-pin
- Programmable Main Power or Standby Power Functionality
- Standard Baud Rates to 115.2 Kbps, Custom Baud Rates to 1.5 Mbps

• Programmable Timer Interface
- Two16-bit Auto-reloading Timer Instances
  - 16 bit Pre-Scale divider
  - Halt and Reload control
  - Auto Reload
- Two 32-bit Auto-reloading Timer Instances
  - 16 bit Pre-Scale divider
  - Halt and Reload control
  - Auto Reload
- Three Operating Modes per Instance: Timer (Reload or Free-Running) or One-shot.
  - Event Mode is not supported

• 32-bit RTOS Timer
- Runs Off 32kHz Clock Source
- Continues Counting in all the Chip Sleep States regardless of Processor Sleep State
- Counter is Halted when Embedded Controller is Halted (e.g., JTAG debugger active, break points)
- Generates wake-capable interrupt event

• Watch Dog Timer (WDT)
- Watchdog reset IRQ vector

• Embedded Reset Engine
- Resets the EC if external VCI_IN0# pin is held low for a programmed time

• 9 Programmable Pulse Width Modulator (PWM) outputs
- Multiple Clock Rates
- 16-Bit ON & 16-Bit OFF Counters

• 4 Fan Tachometer Inputs
- 16 Bit Resolution

• Two RPM-Based Fan Speed Controllers
- Each includes one Tach input and one PWM output
- Each includes one Tach input and one PWM output
- 3% accurate from 500 RPM to 16k RPM
- Automatic Tachometer feedback
- Aging Fan or Invalid Drive Detection
- Spin Up Routine
- Ramp Rate Control
- RPM based Fan Control Algorithm

• Breathing LED Interface
- 3 Blinking/Breathing LEDs
- Programmable Blink Rates
- Piecewise Linear Breathing LED Output Controller
  - Provides for programmable rise and fall waveforms
- Operational in EC Sleep States

• Optional support for Physically Unclonable Function (PUF)
- 2K Byte memory reserved for PUF.

• PS2 Controller
- Two PS2 controllers
- Three PS2 ports
- Both ports are 5 volt tolerant

• PROCHOT interface with Two instances of the PowerGuard Technology
- Monitor for single assertions or cumulative PROCHOT active time
- Interrupt generation for PROCHOT assertion events
- Support PROCHOT assertions to external CPU
- PowerGuard Technology monitors total system power via dedicated Fast A/D converter
- Two programmable thresholds with hysteresis and filtering for each V_ISYS input
- Integrated with PROCHOT interface to provide CPU throttling
- Fast programmable response on high threshold
- Programmable delayed response on low threshold

• Microchip BC-Link Interconnection Bus

- One High/Low speed Bus master controllers

Analog Features

- ADC Interface
  - 10-bit or 12-bit readings supported
  - ADC Conversion time 0.625us/channel
  - Up to 16 Channels
  - External voltage reference
  - Supports thermistor temperature readings
- Two Analog Comparators
  - May be used for Hardware Shutdown
    - Detection of voltage limit event
    - Detection of Thermistor Over-Temp Event

Battery Powered Peripherals

- Real Time Clock (RTC)
  - VBAT Powered
  - 32KHz Crystal Oscillator or External single-ended 32 kHz clock source
  - Time-of-Day and Calendar Registers
  - Programmable Alarms
  - Supports Leap Year and Daylight Savings Time
- Hibernation Timer Interface
  - Two 32.768 KHz Driven Timers
  - Programmable Wake-up from 0.5ms to 128 Minutes
- Week Timer
  - System Power Present Input Pin
    - Week Alarm Event only generated when System Power is Available
  - Power-up Event
  - Week Alarm Interrupt with 1 Second to 8.5 Year Time-out
  - Sub-Week Alarm Interrupt with 0.50 Seconds - 72.67 hours time-out
  - 1 Second and Sub-second Interrupts
- VBAT-Powered Control Interface (VCI)
  - 3 or 4 Active-low VCI Inputs
  - 1 Active-high VCI Output Pin
  - System Power Present Detection for gating RTC wake events
  - Optional filter and latching
- 3 Battery-Powered General Purpose Output (BGPO) Pins

Debug Features

- 2-pin Serial Wire Debug (SWD) interface
- 4-Pin JTAG interface for Boundary Scan
- 1-Pin ITM interface
- Trace FIFO Debug Port (TFDP)

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1.0 GENERAL DESCRIPTION

The MEC165xB is a family of low power integrated embedded controller designed for notebook applications storage enclosure platforms. The MEC165xB is a highly-configurable, mixed-signal, advanced I/O controller. It contains a 32-bit ARM® Cortex-M4 processor core with closely-coupled memory for optimal code execution and data access. An internal ROM, embedded in the design, is used to store the power on/boot sequence and APIs available during run time. When VTR_CORE is applied to the device, the secure bootloader API is used to download the custom firmware image from the system's shared SPI Flash device, thereby allowing system designers to customize the device's behavior.

The MEC165xB device is directly powered by a minimum of two separate suspend supply planes (VBAT and VTR) and senses a third runtime power plane (VCC) to provide "instant on" and system power management functions. The MEC165xB has one banks of I/O pins that are able to operate at 3.3 V (VTR1), one bank that is 1.8V (VTR3) and one bank that can operate at 3.3V/1.8V (VTR2). Operating at 1.8V allows the MEC165xB to interface with the latest platform controller hubs and will lower the overall power consumed by the device, Whereas 3.3V allows this device to be integrated into legacy platforms that require 3.3V operation.

The MEC165xB host interface is the Intel® Enhanced Serial Peripheral Interface (eSPI). The eSPI Interface is a 1.8V interface that operates in single, double and quad I/O modes. The eSPI Interface supports all four eSPI channels: Peripheral Channel, Virtual Wires Channel, OOB Message Channel, and Run-time Flash Access Channel. The eSPI hardware Flash Access Channel is used by the Boot ROM to support Controller Attached Flash Sharing (CAFS). In addition, the MEC165xB has specially designed hardware to support Target Attached Flash Sharing (TAFS). The eSPI TAFS Bridge imposes Region-Based Protection and Locking security feature, which limits access to certain regions of the flash to specific hosts. There may be one or more hosts (e.g., BIOS, ME, etc) that will access the SAF via the eSPI interface. The ARM® Cortex-M4 processor is also considered a host, which will also have its access limited to EC only regions of SPI Flash as determined by the customer firmware application.

The MEC165xB secure bootloader authenticates and optionally decrypts the SPI Flash OEM boot image using the AES-256, ECDSA, SHA-512 cryptographic hardware accelerators. The MEC165xB hardware accelerators support 128-bit and 256-bit AES encryption, ECDSA and EC_KCDSA signing algorithms, 1024-bits to 4096-bits RSA and Elliptic asymmetric public key algorithms, and a True Random Number Generator (TRNG). Runtime APIs are provided in the ROM for customer application code to use the cryptographic hardware. Additionally, the device offers lockable OTP storage for private keys and IDs.

The MEC165xB is designed to be incorporated into low power PC architecture designs and supports ACPI sleep states (S0-S5). During normal operation, the hardware always operates in the lowest power state for a given configuration. The chip power management logic offers two low power states: light sleep and heavy sleep. These features can be used to support S0 Connected Standby state and the lower ACPI S3-S5 system sleep states. In connected standby, any eSPI command will wake the device and be processed. When the chip is sleeping, it has many wake events that can be configured to return the device to normal operation. Some examples of supported wake events are PS2 wake events, RTC, Week Alarm, Hibernation Timer, or any GPIO pin.

The MEC165xB offers a software development system interface that includes a Trace FIFO Debug port, a host accessible serial debug port with a 16C550A register interface, a Port 80 BIOS Debug Port, and a 2-pin Serial Wire Debug (SWD) interface. Also included is a 4-wire JTAG interface used for Boundary Scan testing.

The MEC165xB also supports eSPI host interface, with Controller Attached Flash and Target Attached Flash Sharing in the 128 pin and 144 pin package.

1.1 Family Features

TABLE 1-1: MEC165XB FEATURE LIST

MEC165xB Product Features	MEC1653B-B0-I/TF	MEC1657B-B0-I/TF	MEC1653B-B0-I/SZ	MEC1657B-B0-I/SZ
Device ID	0x002D63XX	0x002DE3XX	0x002D64XX	0x002DE4XX
JTAG ID	0x022D2445	0x022D2445	0x022D2445	0x022D2445
Package	128 WFBGA	128 WFBGA	144 WFBGA	144 WFBGA
Total SRAM Options	416KB	416KB	416KB	416KB

MEC165xB Product Features	MEC1653B-B0-I/TF	MEC1657B-B0-I/TF	MEC1653B-B0-I/SZ	MEC1657B-B0-I/SZ
Code/Data Options (Primary Use)	352KB/64KB	352KB/64KB	352KB/64KB	352KB/64KB
Battery Backed SRAM	128 bytes	128 bytes	128 bytes	128 bytes
Clock Speed	96MHz	96MHz	96MHz	96MHz
MPU	Yes	Yes	Yes	Yes
FPU	No	No	No	No
EEPROM Controller Supports 8KB	Yes	No	Yes	No
Internal SPI	No	Yes	No	Yes
USB 2.0	No	No	No	No
Optional USB 1.1	No	No	No	No
2 pin SWD	Yes	Yes	Yes	Yes
4 pin JTAG	Yes	Yes	Yes	Yes
eSPI Host Interface	Yes	Yes	Yes	Yes
eSPI SAF Interface	Yes	Yes	Yes	Yes
GPIO Support through eSPI Virtual Wire	Yes	Yes	Yes	Yes
RPMC	Yes	Yes	Yes	Yes
SPI Target (Single Wire Mode)	Yes	Yes	Yes	Yes
GP-SPI	Yes	Yes	Yes	Yes
I3C Host Controller	1	1	1	1
I3C Secondary Controller	1	1	1	1
SMBus Network 2.0/ I2C Host Controllers	5	5	5	5
SMBus Ports	15	15	16	16
8042 Emulated Keyboard Controller	Yes	Yes	Yes	Yes
Embedded Memory Interface (EMI)	3	3	3	3
Mailbox Register Interface	1	1	1	1
ACPI Embedded Memory Controller Interface	5	5	5	5
ACPI PM1 Block Interface	1	1	1	1
Parallel Port	No	No	No	No
Trace FIFO Debug Port	Yes	Yes	Yes	Yes
Internal DMA Channels	20	20	20	20
16-bit Basic Timer	4	4	4	4
32-bit Basic Timer	2	2	2	2
16-bit Counter/Timer	4	4	4	4
Capture Timer	6	6	6	6
ICT Channels	13	13	16	16
Compare Timer	2	2	2	2
Watchdog Timer (WDT)	1	1	1	1

MEC165xB Product Features	MEC1653B-B0-I/TF	MEC1657B-B0-I/TF	MEC1653B-B0-I/SZ	MEC1657B-B0-I/SZ
Hibernation Timer	2	2	2	2
Week Timer	1	1	1	1
Sub Week Timer	1	1	1	1
RTC	1	1	1	1
RTOS Timer	1	1	1	1
Dedicated Battery-Powered General Purpose Output (BGPO)	1	1	1	1
BGPO Multiplexed with GPIO's	2	2	2	2
Active Low VBAT-Powered Control Interface (VCI)	3	3	4	4
Active high VBAT-Powered Control Interface (VCI_OVRD_IN)	1	1	1	1
VCI_OUT	1	1	1	1
2 Pin Parallel XTAL Oscillator	Yes	Yes	Yes	Yes
Keyboard Matrix Scan Support	Yes	Yes	Yes	Yes
Port 80 BIOS Debug Port	2	2	2	2
PECI 3.1 Interface	Yes	Yes	Yes	Yes
PS/2 Device Interface	2 controllers/ 3 ports	2 controllers/ 3 ports	2 controllers/ 3 ports	2 controllers/ 3 ports
GPIOs	108	108	123	123
Blinking/Breathing LED	3	3	3	3
General Purpose SPI Host Controller	0	0	0	0
Quad SPI Host Controller	1 controller/ 3 ports	1 controller/ 3 ports	1 controller/ 3 port	1 controller/ 3 port
10/12-bit ADC Channels	8	8	8	8
Vref-2 ADC	Yes	Yes	Yes	Yes
Analog Comparator	2	2	2	2
RPM2PWM	2	2	2	2
16-bit PWMs	9	9	11	11
16-bit TACHs	4	4	8	8
UARTs	3 UART0: 8-pin UART1: 4-pin UART2: 8-pin	3 UART0: 8-pin UART1: 4-pin UART2: 8-pin	3 UART0: 8-pin UART1: 8-pin UART2: 8-pin	3 UART0: 8-pin UART1: 8-pin UART2: 8-pin
AES Hardware Support	128-256 bit	128-256 bit	128-256 bit	128-256 bit
PQC Support •LMS Secure Boot Parameter Set: -LMOTS_SHA256_N24_W4 -LMS_SHA256_M24_H20 • ML-DSA-87 (Crystals Dilithium-5)	Yes	Yes	Yes	Yes
SHA 1, SHA 2 and Hashing Support	SHA-1 to SHA-2	SHA-1 to SHA-2	SHA-1 to SHA-2	SHA-1 to SHA-2

MEC165xB Product Features	MEC1653B-B0-I/TF	MEC1657B-B0-I/TF	MEC1653B-B0-I/SZ	MEC1657B-B0-I/SZ
Public Key Cryptography Support	RSA: 4K bit ECC: 571 bit	RSA: 4K bit ECC: 571 bit	RSA: 4K bit ECC: 571 bit	RSA: 4K bit ECC: 571 bit
True Random Number Generator with health test	1K bit	1K bit	1K bit	1K bit
User OTP	4K bits	4K bits	4K bits	4K bits
5V Tolerant Pads	15	15	15	15
GPIO Pass Through Ports (GPTP)	6	6	8	8
BC-Link	0	0	0	0
Prochot Monitor input only (Optional)	1	1	1	1
Prochot Monitor IO (bidi pin) (Optional)	1	1	1	1
Power Guard (Optional)	2	2	2	2
RC-ID	3	3	3	3
Differential Power Analysis countermeasures (DPA)	No	No	No	No
DICE	No	No	No	No
Root Of Trust	Yes	Yes	Yes	Yes
Secure Boot	Yes	Yes	Yes	Yes
Immutable Code	Yes	Yes	Yes	Yes
Key Revocation	Yes	Yes	Yes	Yes
Key Roll Back Protection	127	127	127	127
Transfer Of Ownership	Yes	Yes	Yes	Yes
Crisis Recovery	Yes	Yes	Yes	Yes
Optional PUF Support	Yes	Yes	Yes	Yes
Boots From	TAFS/CAFS/G3	TAFS/CAFS/G3	TAFS/CAFS/G3/Internal SPI Flash	TAFS/CAFS/G3/Internal SPI Flash

Note 1: Please refer to Boot ROM document for optional OTP selectable features.

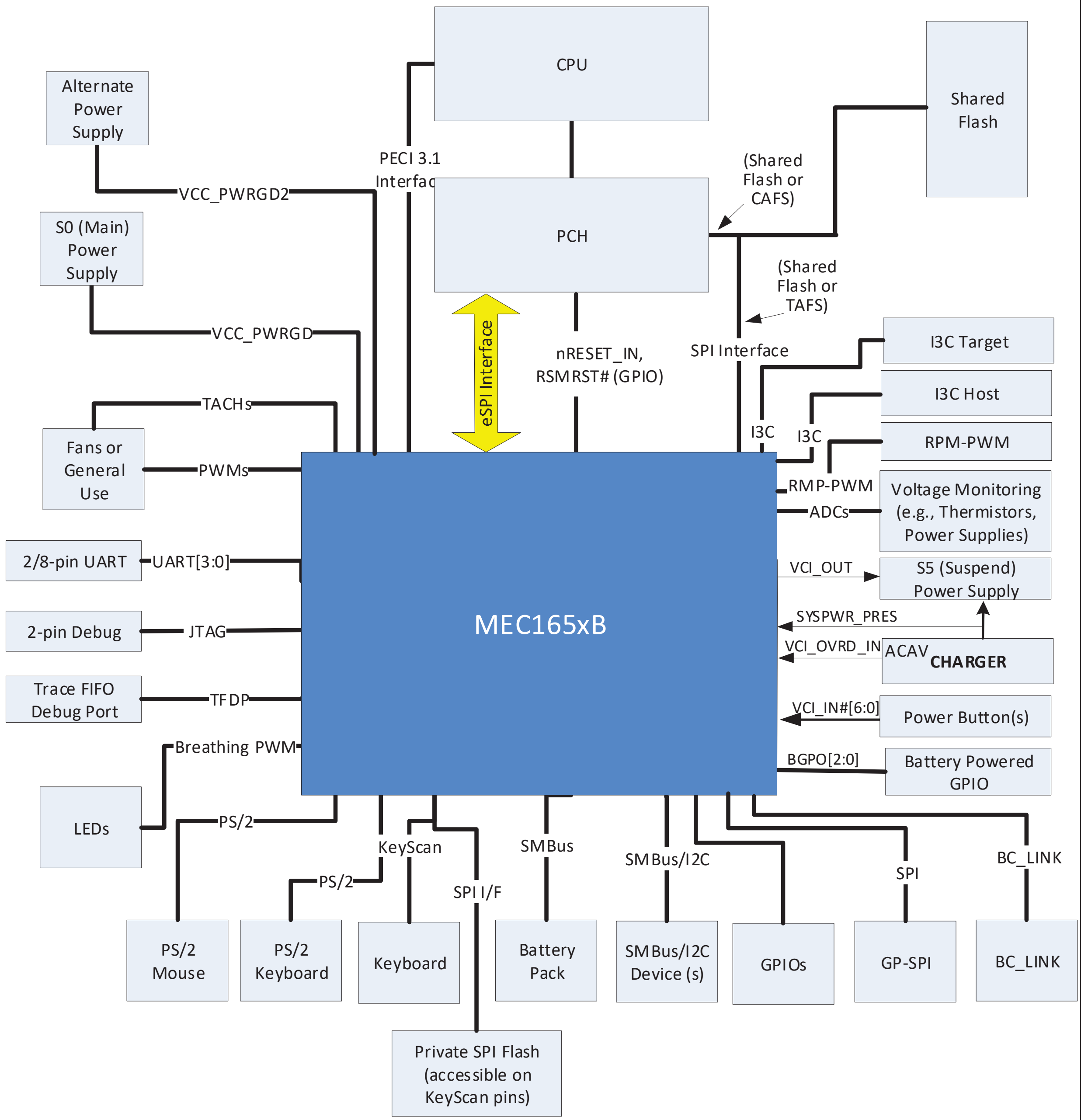
1.2 Boot ROM

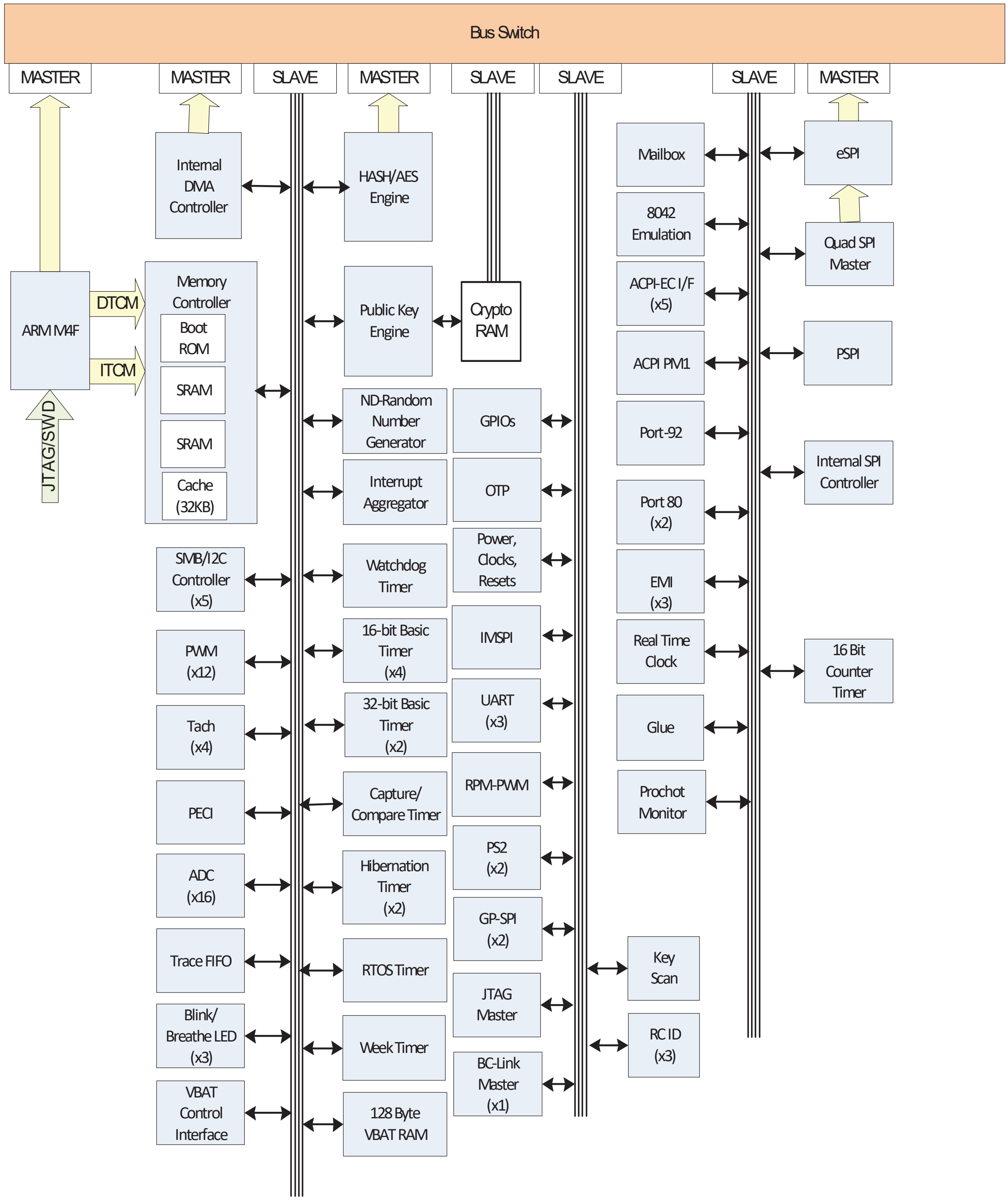
Following the release of the RESET_EC signal, the processor will start executing code from the Boot ROM. The Boot ROM executes the SPI Flash Loader, which downloads User Code from SPI Flash and stores it in the internal Code RAM. Refer to MEC165xB Boot ROM document for further details.

1.3 MEC165xB Internal Address Spaces

The Internal Embedded Controller can access any register in the EC Address Space or Host Address Space. If the I^2C interface is used as the Host Interface, access to all the IP Peripherals is dependent on EC firmware.

FIGURE 1-1: BLOCK DIAGRAM




Bus Switch
MASTER
MASTER
SLAVE
MASTER
SLAVE
SLAVE
SLAVE
MASTER
ARM M4F
DTCM
ITCM
JTAG/SWD
Internal DMA Controller
Memory Controller
Boot ROM
SRAM
SRAM
Cache (32KB)
SMB/I2C Controller (x5)
PWM (x12)
Tach (x4)
PECI
ADC (x16)
Trace FIFO
Blink/ Breathe LED (x3)
VBAT Control Interface
HASH/AES Engine
Public Key Engine
Crypto RAM
ND-Random Number Generator
Interrupt Aggregator
Watchdog Timer
16-bit Basic Timer (x4)
32-bit Basic Timer (x2)
Capture/ Compare Timer
Hibernation Timer (x2)
RTOS Timer
Week Timer
128 Byte VBAT RAM
GPIOs
OTP
Power, Clocks, Resets
IMSPI
UART (x3)
RPM-PWM
PS2 (x2)
GP-SPI (x2)
JTAG Master
BC-Link Master (x1)
Mailbox
8042 Emulation
ACPI-EC I/F (x5)
ACPI PM1
Port-92
Port 80 (x2)
EMI (x3)
Real Time Clock
Glue
Prochot Monitor
Key Scan
RC ID (x3)
eSPI
Quad SPI Master
PSPI
Internal SPI Controller
16 Bit Counter Timer